

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EX-402 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Electrical and Electronics Materials
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.

ii) All questions carry equal marks.

1. A coil of copper wire has a resistance of 50 ohms at 20°C. Calculate its resistance at 60°C. Given that temperature. Co-efficient of 0°C is 0.00427 per °C.
2. What is Super conductivity? Name some super conducting materials. What are the properties of super conducting materials.
3. Mention the yield of application of the following insulating materials with reasons against each .
 - i) Fibre glass
 - ii) Mica
 - iii) Ceramics
4. What are the thermocouples and what are their applications? Name some materials which are used for the construction of thermocouples.
5. a) Classify semiconductors. Also list the factors which affect the resistivity of semiconductor.
b) What is P-N diode? Explain the common techniques used for the fabrication of P-N diode.
- OR
- a) What is meant by IC packaging? Discuss characteristics of IC component.
- b) Compare fabrication process of BJT and FET.
6. Write the characteristics and applications of
 - i) LED
 - ii) Photodiode
7. Discuss the applications of semiconductors to obtain non-linear resistors and temperature sensitive resistors.

OR

The critical field for Niobium is 1×10^5 Amp./m. at 8 K and 2×10^5 Amp./m. at 0 K. Calculate the critical temperature of the material.

8. Write short notes on: (any two)
 - a) Soft and hard magnetic materials
 - b) Fuel cells
 - c) Properties of ceramic and uses
 - d) Light dependent resistors
 - e) Underground cables
